

Quick Start

1. Course Content

Complete the initial configuration of the MicroROS-V2 development environment and be able to control the robot's movement with a gamepad or keyboard. This chapter covers:

1. Starting the MicroROS_V2_VM virtual machine
2. Configuring the robot Wi-Fi and micro-ROS Agent parameters
3. Gamepad control (USB Host direct control / ROS2 topic control)
4. Keyboard control

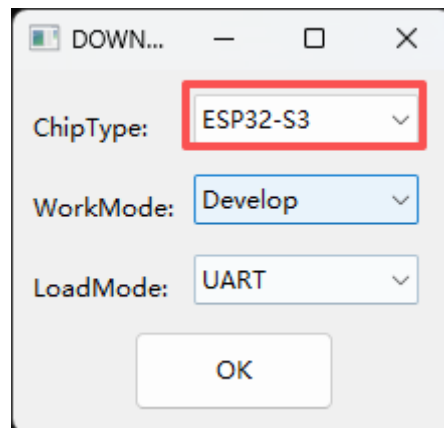
2. Preparation

2.1 Flashing the Firmware

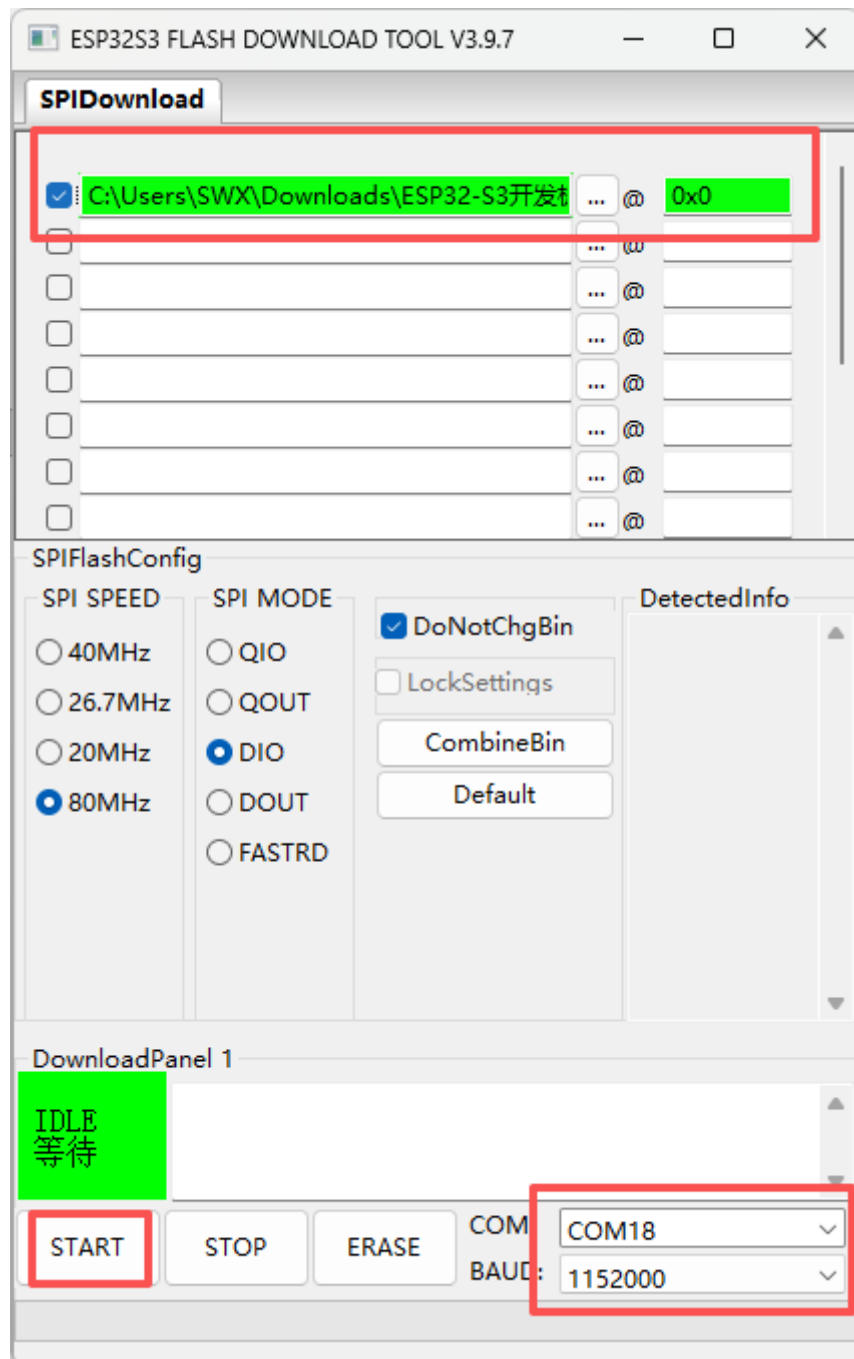
Note: The firmware is already flashed at the factory, so this step can be skipped.

Use the flash_download_tool utility to flash the factory firmware (MicroROS-V2-v1.1.0-complete.bin) to the MicroROS-V2 control board (ESP32-S3).

Connect the control board to the computer with a USB data cable, open the flash_download_tool software, and flash the factory firmware.



After configuring as shown in the image, click **START** to begin flashing. Set the firmware path according to your local actual path, and the port likewise.



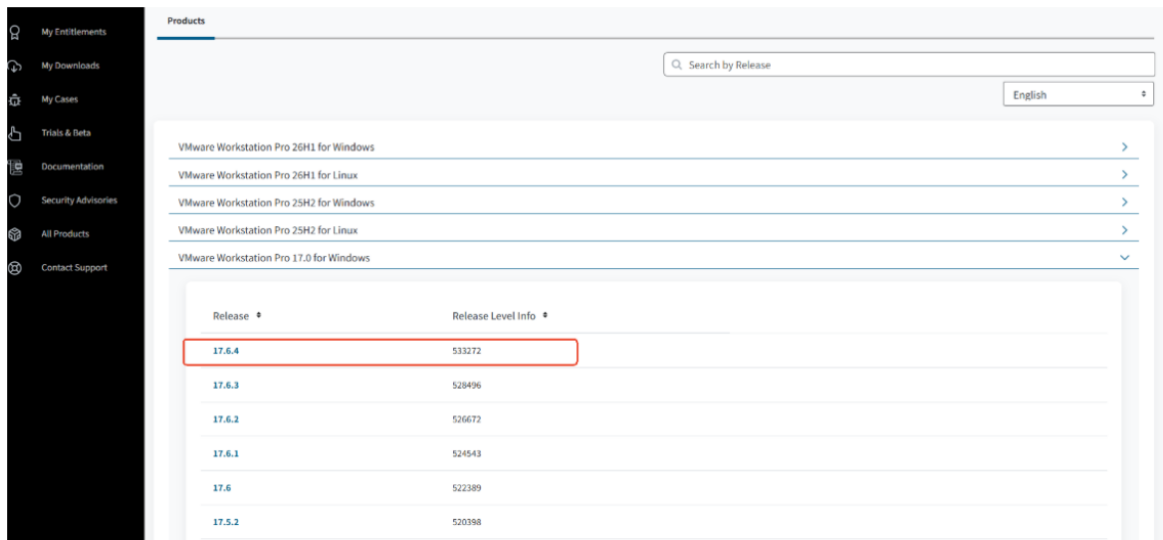
During flashing, wait for the progress bar to finish. When **FINISH** is displayed, the flashing is successful. At this point you can reset the main control board and the firmware will take effect.

2.2 Opening the Virtual Machine

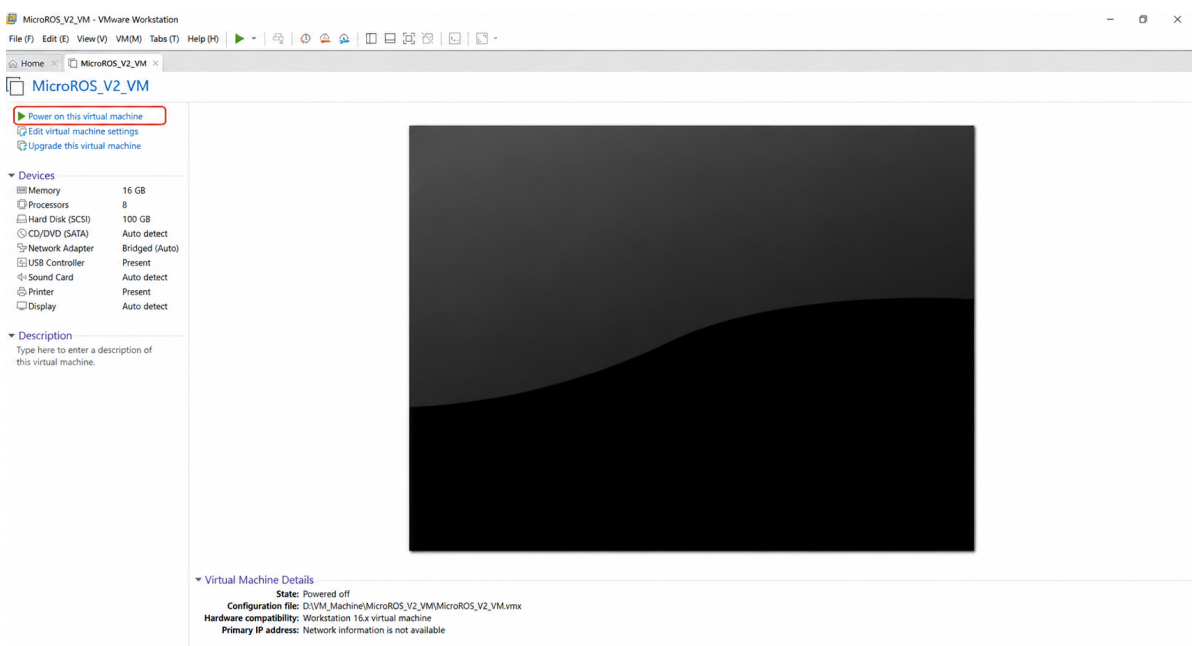
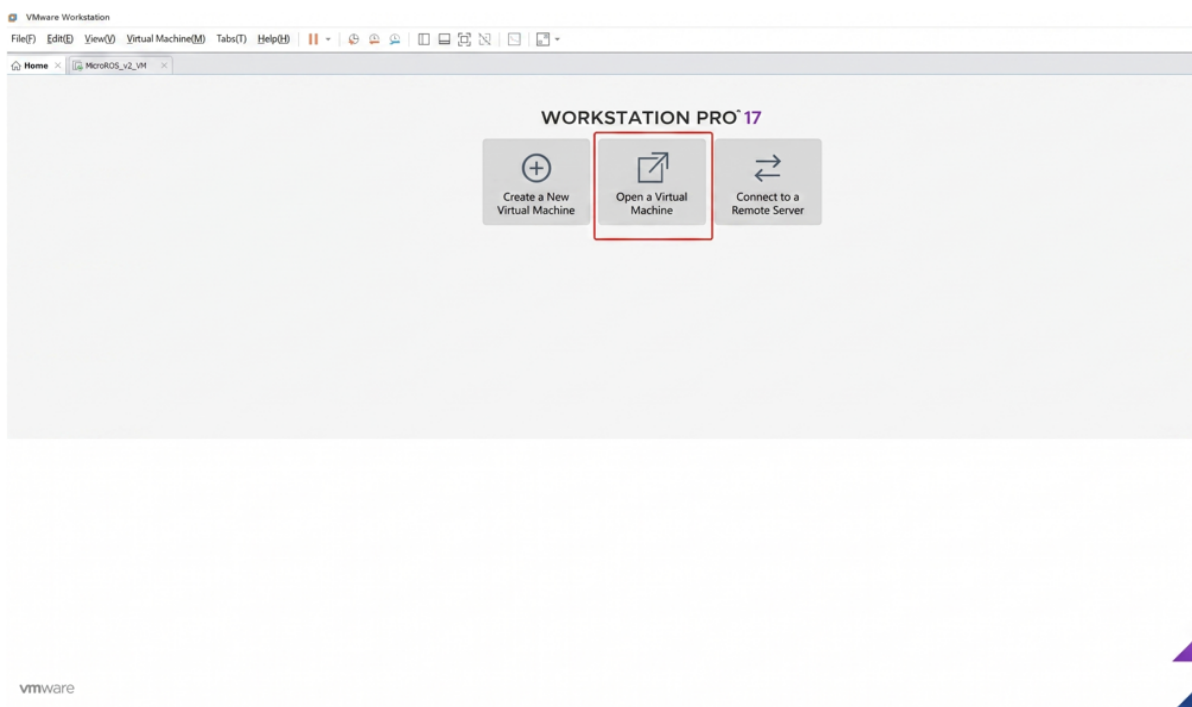
The VMware platform can be installed by yourself.

Download address (the address may change; please refer to the official VMware website):

<https://support.broadcom.com/group/ecx/productdownloads?subfamily=VMware%20Workstation%20Pro&freeDownloads=true>



Start the MicroROS-V2 virtual machine environment



2.3 Control Board Parameter Configuration

1. Modifying the Configuration Parameters

Modify the parameters in the custom mode in `~/workspaces/tools/config_robot.py` according to your actual network.

Users should first confirm the Wi-Fi, Agent IP, Agent port, and ROS_DOMAIN_ID. Other parameters can remain at their default values and only need to be modified when calibrating the gimbal, adjusting the control effect, or configuring the camera.

Parameter	Default Value	Description	Modification Suggestion
main_wifi_ssid	SSID	Wi-Fi name that the robot's main controller connects to	Change to an available 2.4G Wi-Fi; the ESP32-S3 does not support 5G Wi-Fi
main_wifi_passwd	PASSWORD	Wi-Fi password that the robot's main controller connects to	Must match the Wi-Fi name above
agent_ip_addr	[192, 168, 33, 146]	IP of the computer/virtual machine running the micro-ROS Agent	Change to the real IP of the virtual machine or host on the same LAN; keep the format of 4 numbers
agent_ip_port	8090	micro-ROS Agent UDP port	No need to change by default; if you specify a different port when starting the Agent, keep it consistent here
ros_domain_id	77	ROS_DOMAIN_ID	Must be consistent with the ROS_DOMAIN_ID in the virtual machine terminal
ros_namespace	""	ROS namespace	Keep an empty string for a single robot; fill it in when multiple robots need to distinguish topics, e.g. "robot1"
servo_1_offset	0	S1 horizontal gimbal offset, in degrees	Fine-tune when the camera is not facing straight ahead; positive value turns right, negative value turns left
servo_2_offset	0	S2 tilt gimbal offset, in degrees	Fine-tune when the camera tilt angle deviates; positive value tilts up, negative value tilts down
motor_pid	(1.8, 0.12, 0.0)	Chassis motor speed-loop PID	Do not modify for ordinary users; only change when the motion control effect is abnormal and tuning is clearly needed
imu_yaw_pid	(2.4, 0.0, 0.16)	Straight-line heading-hold PID	Do not modify for ordinary users; only change when the straight-line holding effect needs optimization

2. Servo Offset Adjustment Instructions

- **servo_1_offset** adjusts the horizontal gimbal S1. When the camera is facing too far to the left, appropriately increase the value; when the camera is facing too far to the right, appropriately decrease the value.
- **servo_2_offset** adjusts the tilt gimbal S2. When the camera is facing too low, appropriately increase the value; when the camera is facing too high, appropriately decrease the value.
- It is recommended to make small adjustments of $1^{\circ} \sim 3^{\circ}$ each time, and then re-run `python3 ~/workspaces/tools/config_robot.py custom` to observe the effect.
- If the actual assembly direction is opposite to the above, take the actual motion direction as the standard and adjust the offset value in the opposite direction.

3. Parameter Configuration

1. Turn on the robot's power, connect the robot control board to the virtual machine with a Type-C data cable, and then run the configuration command:

```
python3 ~/workspaces/tools/config_robot.py custom
```

```
yahboom@yahboom:~$ python3 ~/workspaces/tools/config_robot.py custom
Configuring...

=====
current firmware info
=====
Yahboom
Name: MicroROS-V2
firmware_version: 1.0.0
wifi_name: NX30PRO_2.4G
camera_ip: 192.168.33.154
agent_ip_addr: 192.168.33.146
agent_ip_port: 8090
ros_domain_id: 77
ros_namespace: (none)
servo_1_offset: 5
servo_2_offset: 0
motor_pid: 1.80, 0.12, 0.00
imu_yaw_pid: 2.40, 0.00, 0.16
=====

Configuration complete! Parameters take effect immediately.
```

After the configuration is successful, the robot will start with the new Wi-Fi and Agent parameters. If the ROS indicator light stays on, it means the micro-ROS Agent has been connected.

2. If you need to restore the factory default parameters, run:

```
python3 ~/workspaces/tools/config_robot.py default
```

```

yahboom@yahboom:~/workspaces/MicroROS-V2-FW/robot_base_control/MicroROS-V2-Base-Control-FW/tools$ python3 config_robot.py default
Factory reset issued. Waiting for device to reboot...

=====
current firmware info
=====
Yahboom
Name: MicroROS-V2
firmware_version: 1.0.0
wifi_name: NX30PRO_2.4G
camera_ip: (none)
agent_ip_addr: 192.168.33.146
agent_ip_port: 8090
ros_domain_id: 77
ros_namespace: (none)
servo_1_offset: 0
servo_2_offset: 0
motor_pid: 1.80, 0.12, 0.00
imu_yaw_pid: 2.40, 0.00, 0.16
=====

Factory defaults restored. Parameters take effect immediately.

```

3.1 Handle Button Operations

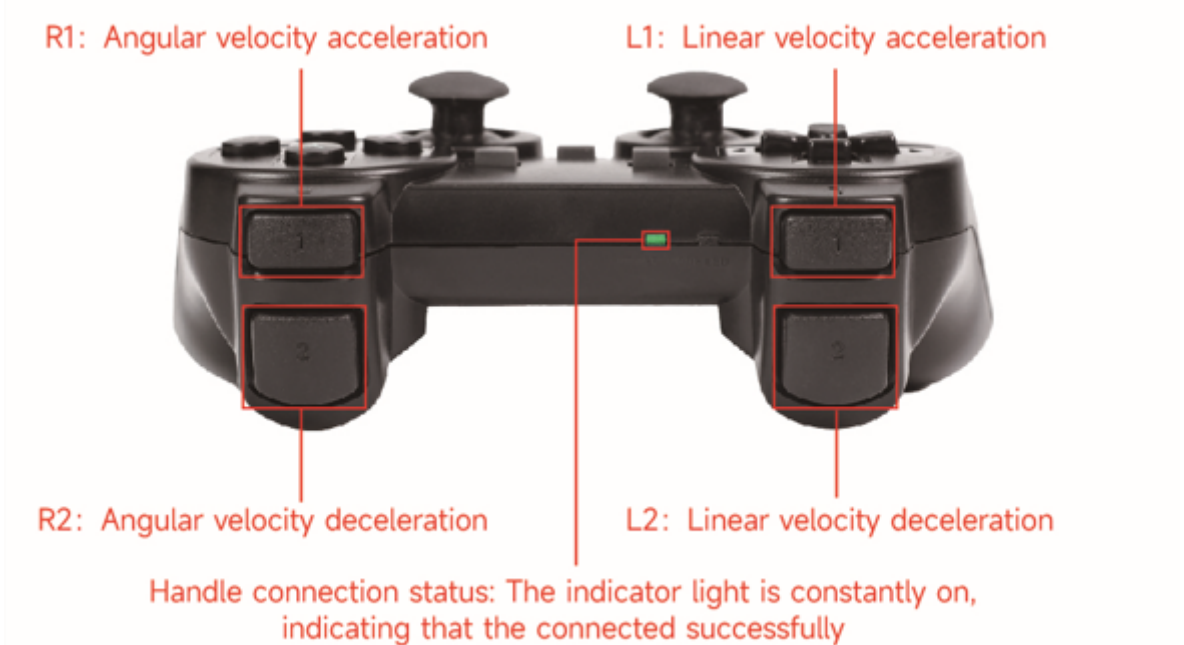
SELECT (Only valid for ROS2 version of the handle control program)
Single press: Enable/disable handle ROS2 program control

Arrow keys
Up/Down: Forward/Backward
Left/Right: Left/Right

START: Press the
buzzer to sound,
Release the buzzer
no sound

(For robot car with PTZ)
Y: PTZ upwards
A: PTZ downwards
X: PTZ turn left
B: PTZ turn right





3.2 Running the Program for Gamepad Control

Quick start:

```
bash ~/workspaces/scripts/start_remote_control.sh joy
```

```
remote_ctrl@bash - "yahboom"
remote_ctrl@bash - "yahboom" 203x32
[esp32_ip_camera_node-2] [INFO] [1782804858.400816889] [esp32_ip_camera_publisher]: publish...
[esp32_ip_camera_node-2] [INFO] [1782804858.401226162] [esp32_ip_camera_publisher]: resize_wi
dth: 320, resize_height: 240
[esp32_ip_camera_node-2] [INFO] [1782804858.401754511] [esp32_ip_camera_publisher]: enable_ra
w_publish: False, raw_topic: /camera/image_raw
[esp32_ip_camera_node-2] [INFO] [1782804858.402115256] [esp32_ip_camera_publisher]: enable_co
mpressed: True, compressed_topic: /camera/image_raw/compressed, jpeg_quality: 15
[esp32_ip_camera_node-2] [INFO] [1782804858.402489215] [esp32_ip_camera_publisher]: startup_d
elay: 3.0s, reconnect: 1.0s - 30.0s (exponential backoff)
[esp32_ip_camera_node-2] [WARN] [1782804861.383015154] [esp32_ip_camera_publisher]: Camera no
t opened, trying reconnect... (next retry in 1.0s)
[esp32_ip_camera_node-2] [INFO] [1782804861.522296844] [esp32_ip_camera_publisher]: IP camera
stream opened successfully

LOCAL_IP: 192.168.33.146
ROS_DOMAIN_ID: 77
yahboom@yahboom:~$
yahboom@yahboom:~$ source /home/yahboom/workspaces/MicroROS-V2-ROS2-SRC/microros_car_v2_ws/install/setup.bash
yahboom@yahboom:~$ ros2 launch microros_v2_remote_operation_pkg microros_v2_joy_control.launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-06-30-15-34-20-436098-yahboom-7-05296
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [joy_node-1]: process started with pid [785316]
[Joy_node-1] [INFO] [1782804860.991688696] [joy_node]: No haptic (rumble) available, skipping initialization
[Joy_node-1] [INFO] [1782804860.991849088] [joy_node]: Opened joystick: Controller. deadzone: 0.850000
[INFO] [joy_driver_node-2]: process started with pid [785349]
[INFO] [joy_control_signal_node-3]: process started with pid [785351]
[INFO] [microros_v2_usb_joy_control_node-4]: process started with pid [785353]
[Joy_control_signal_node-3] [INFO] [1782804861.803250047] [joy_control_signal_node]: joy_control_signal_node
started: /joy_data -> /joy_car_control
[Joy_driver_node-2] [INFO] [1782804861.982059152] [joy_driver_node]: joy_driver_node started: /joy -> /joy_da
ta @ 50Hz
[microros_v2_usb_joy_control_node-4] [INFO] [1782804861.983305052] [microros_v2_usb_joy_control_node]: micror
os_v2_usb_joy_control_node started: /joy_car_control -> /cmd_vel, /cmd_ptz_angle, /beep
[microros_v2_usb_joy_control_node-4] [INFO] [1782804861.983980050] [microros_v2_usb_joy_control_node]: speed
gears: linear(G2)=0.30 m/s, angular(G2)=1.50 rad/s; soft-stop: lin=0.80 m/s², ang=2.00 rad/s²; servo: period=
0.010s, speed=180.0 dps, accel=360.0 dps²
```

The script will automatically start the state estimation layer, the gamepad driver layer, and the remote control operation layer.

3.3 Motion Control

An independent per-axis arbitration between the joysticks and the D-pad is used — the joystick takes priority, and the D-pad only acts as a backup when the corresponding joystick axis is idle:

Input	Function	Curve	Description
Left joystick up/down (LY)	Forward / Backward	Square curve (power=2.0)	Analog control, with a 0.15 dead zone
Right joystick left/right (RX)	Turn left / Turn right	Cubic curve (power=3.0)	Analog control, with a 0.15 dead zone
Left joystick left/right (LX)	No function	—	Not used
Right joystick up/down (RY)	No function	—	Not used
D-pad up/down	Forward / Backward	—	Effective only when LY is idle (digital backup)
D-pad left/right	Turn left / Turn right	—	Effective only when RX is idle (digital backup)

3.4 Speed Gears

There are 5 fixed speed gears in total (G1–G5), with G2 as the default:

Gear	Linear Speed	Angular Speed
G1	0.20 m/s	0.50 rad/s
G2 ★	0.30 m/s	1.50 rad/s
G3	0.50 m/s	2.50 rad/s
G4	0.75 m/s	3.75 rad/s
G5	1.00 m/s	5.00 rad/s

Button	Function
L1	Increase linear speed by one gear
L2	Decrease linear speed by one gear
R1	Increase angular speed by one gear
R2	Decrease angular speed by one gear

Raising the gear to G5 or lowering it to G1 triggers a double beep notification.

3.5 Handle Control

Button	Function	Description
X	Servo S1 forward	Effective when X is pressed alone
B	Servo S1 reverse	Effective when B is pressed alone
Y	Servo S2 reverse	Effective when Y is pressed alone
A	Servo S2 forward	Effective when A is pressed alone

S1 travel range: -90° ~ +90°; S2 travel range: -90° ~ +20°.

3.6 Buzzer and Enable Switch

Button	Function
START (OPTIONS)	Hold to keep beeping, release to stop
SELECT (SHARE)	Toggle the gamepad enable state (<code>/JoyState</code> : True ↔ False)